

Nebico Oil Retention – PLC Overview

As-Built Notes on the Ladder Logic

Nebico Oil Retention Project – Program 3.3 (Coolmay L10S)

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Location	Kathmandu, Nepal
Prepared	09 September 2026
This edition	15 September 2026
Document	PLC Guide, Program 3.3 – As-Built Edition
Source	Coolmay L10S ladder export, “MAIN” project, 15-Sep-2026 print
Supersedes	PLC Guide, Program 3.2 – As-Built Edition
HMI	TK8100H × 1 (supervisory), TK8050H × 3 (per-station)

This is my own working guide to Program 3.3, exported from a Coolmay L10S CPU. The same four changes named on the programme’s own cover carry over from Program 3.2: pointer functions to shorten bulk calculation, a revised core (Mode Scheduler) logic, revised Oil Distribution, and a general clean-up of segmentation. On top of that, the manual tank-disconnect override for tanks A, B and C (Section 6) is unchanged, and this edition adds a fault trigger-value snapshot for HMI display (Section 5.6). Section 2 summarises everything that changed against the 3.2 edition, and the main body below is a full reconstruction from the exported rungs, confirmed against the live programme and the HMI configuration throughout.

Contents

1	Purpose and Scope	2
2	What Changed: 3.2 → 3.3	3
3	Register Bands and the Pointer Mechanism	4
4	Program Structure Overview	6
5	Functional Walkthrough	7
5.1	Timer Chain (Step 0)	7
5.2	Load Once (Step 22) and Declarations (Step 65)	7
5.3	E-Stops (Step 43) and Automation (Step 56)	7
5.4	Analog to Digital, ADC Float, RTD/Ultrasonic/Flowmeter Scaling (Steps 133–336)	7
5.5	Sensor Averaging (Steps 336–404) and Level % Converter (Step 405)	7
5.6	Error Threshold (Step 614)	7
5.7	Water Temperature Target Logic (Step 915)	8
5.8	Heater Latching (Step 1038) and Manual Heater Control (Step 1210)	8
5.9	Height to Mass (Step 1220) and Dynamic Upper Thermal Boundary (Step 1273)	8
5.10	PO Designation (Step 1349), PO Timer/D302 (Step 1410), PO Dispensing (Step 1587)	8
5.11	Tank Scoring and Manual Disconnect Override (Step 1605)	8
5.12	Need Scheduler through Scheduler Gates (Steps 1760–2578)	9
5.13	Solar Gate (Step 2582) and Manual Solar Gate (Step 2607)	9
5.14	Oil Scheduler / MODBUS (Step 2611)	9
5.15	Oil Distribution Calculation, Gating and Timing (Steps 2843–3014)	9
5.16	Jacket Pump (Step 3015), AODD Pump (Step 3020), Manual Pump Mode (Step 3053)	9
5.17	HMI Display for Inputs (Step 3143), Oil Chart (Step 3207), Charting (Step 3277)	10
6	Manual Tank Disconnect Override	10
6.1	How it works	10
6.2	Why this location in the logic	10
7	Timer/Counter Presets	11
8	Reconstructed Register Map	12
9	Calibration / Placeholder Constants	14

1 Purpose and Scope

This is my guide to **PLC Program 3.3** for the Nebico Oil Retention system, exported from a **Coolmay L10S** CPU, superseding the Program 3.2 edition of this same document. I am writing it as an archival, as-built reference: to record how this ladder programme works, to give myself a starting point for future changes, and to document the system-engineering and PLC-programming behind it.

- Section 2 summarises what changed between the 3.2 export and this 3.3 export, at a glance.
- Section 3 sets out the raw/processed/average register bands and the pointer (Z0-Z5) / sub-routine (P0-P5) mechanism.
- Section 4 gives the programme's own line statement index, renumbered for this export.
- Section 5 walks through the programme section by section.
- Section 6 covers the manual tank-disconnect override, unchanged in this export.
- Section 7 and Section 8 carry the timer/counter presets and the reconstructed register map, both updated for 3.3.
- Section 9 lists the constants I have confirmed as site-calibration placeholders.

2 What Changed: 3.2 → 3.3

Same programme, renumbered and lightly extended. Every named section in 3.2 is still present in 3.3 and behaves the same way – but nearly every step number has shifted, by an amount that grows through the programme (roughly +90 near the front, past +130 by the Need Scheduler) rather than a single flat offset, consistent with a handful of small insertions earlier in the programme accumulating. The genuine (not just renumbered) changes in this export:

- **Fault trigger-value snapshot, D750–D768.** New this edition – Section 5.6.
- **PO-dispense and oil-dispense timers renumbered to a 100 ms base.** T5 (PO-melt watchdog) and T100 (oil dispense duration) in 3.2 are now **T250** and **T251**. Both are 100 ms-resolution Coolmay timer devices, the same base already used throughout this ladder (T0's K10 = 1 s is ten 100 ms counts), so their calculated presets are scaled $\times 10$ before loading: D340 = D302 $\times 10$ feeds T250, and D316 is itself multiplied by E10 in place before feeding T251. Section 5.10 and Section 7.
- **A flowmeter-fitted/not-fitted toggle, M800.** An HMI switch selecting whether a physical flowmeter is installed at this site. When set (flowmeter not fitted), the processed flow register D42 is forced to a fixed nominal value of 1, so the Oil Distribution timing maths can still run without a live flow signal. Section 5.15.
- **A master Oil Controller enable, M65, sourced from the TK8100H.** Gates Y005 alongside M11 and the active station's lock – a system-wide dispense enable driven from the supervisory HMI (TK8100H), independent of the three per-station HMIs (TK8050H). Section 5.15.
- **Manual Mode Scheduler merged into Manual Pump Mode.** 3.2 carried these as two separate named sections (steps 1728 and 2899); in 3.3 only *Manual Pump Mode* appears as a named section, at step 3053, and it drives both the manual Y002–Y004 mode-select outputs and the pump outputs.
- **Manual Oil Valve** no longer has its own section header – the rung itself (M12 AND X003 → Y005) is still present, folded into the tail of Oil Distribution Timing around step 3012.

Everything else – register bands, the pointer mechanism, the Tank Scoring weights, the mixing constant, the manual tank-disconnect override, the MODBUS free-for-all lock, the AODD/Jacket pump logic – is functionally identical to 3.2, just at new step numbers.

3 Register Bands and the Pointer Mechanism

Program 3.3 processes the same eleven sensor channels (six temperatures, four levels, one flow) through the same indexed-loop mechanism as 3.2: each loop steps an index register (Z0–Z5) through the channel width and jumps back to its own subroutine pointer (P0–P5) until the index reaches its limit, then falls through.

Band	Registers	Content
Raw	D0–D20	Six raw temperature words (D0, D2 onboard analogue channels; D4–D10 from the Slot 1 RTD module) + four raw ultrasonic level words (D12–D18, Slot 2 module) + one raw flow word (D20, from special register D8231).
Processed temp.	D22–D32	Six RTD channels after linear scaling (P0, “RTD to Temperature”).
Processed level	D34–D40	Four ultrasonic channels after linear scaling (P1, “Ultrasonic to Level”).
Processed flow	D42	Flowmeter after linear scaling (inline, “Flowmeter to RPM”) – or a fixed value of 1 when the M800 not-fitted toggle is set.
Averaging accumulator	D44–D64	Running total per channel, all eleven channels together (P2, “Sensor Averaging”).
Final average	D64–D84	Accumulator divided by 5 and reset (P3); this is the value the rest of the programme reads as “the” sensor value for each channel.
Trigger snapshot	D750–D768	Rate-of-change value captured at the moment a channel’s second (lower) fault threshold is crossed – the HMI “max value recorded” display. New this edition, Section 5.6.

Bands confirmed unchanged from 3.2. The loop limits give the same bands as before: raw 0–20, processed temperature 22–32, processed level 34–40, flow at 42, averaging accumulator 44–64, final average 64–84. Channel order within every band is unchanged: index 0 = Temp T5, 1 = Temp TA, 2 = Temp TB, 3 = Temp TC, 4 = Temp HEA, 5 = Temp SOL, 6 = Level T5, 7 = Level TA, 8 = Level TB, 9 = Level TC, 10 = Flow.

Pointer	Loop limit	Step (subroutine)	Role
Z0	0 to 10 by 2 (6 iter.)	P0, step 223	RTD to Temperature
Z1	0 to 6 by 2 (4 iter.)	P1, step 266	Ultrasonic to Level
Z2	0 to 20 by 2 (11 iter.)	P2, step 336	Sensor Averaging – accumulate (store)
Z3	0 to 20 by 2 (11 iter.)	P3, step 366	Sensor Averaging – divide by 5, reset (compute)
Z4	0 to 20 by 2 (11 iter.)	P4, step 562	Sensor Logging – 5 s snapshot (store)
Z5	0 to 20 by 2 (11 iter.)	P5, step 584	Rate-of-change ratio (compute)

Accumulate rung, confirmed again. The step-336 accumulate rung is still DEADD D22Z2 D44Z2 D44Z2 – new sample added to the existing running total, not doubled – exactly as confirmed for 3.2. P3’s divide is still by 5 (DEDIV D44Z3 E5 D64Z3), so the 5-sample average behaviour is unchanged.

Why the store/compute split, across both pairs. P2/P3 and P4/P5 are each split into a “store” half and a “compute” half rather than combined into one pass – P2 and P4 write the new value into its holding register (the running total, and the 5-second-old snapshot respectively), and P3 and P5 do the actual arithmetic that consumes it (the divide-by-5 average, and the rate-of-change ratio). This is a deliberate authoring choice, not an artefact of the export: kept as separate subroutines because it is easier to track and implement that way while writing the ladder, even though a combined single pass would be marginally more compact.

4 Program Structure Overview

The programme’s own line-statement index for this export, MAIN project, 15-Sep-2026 print, with the bracketed pointer sections ([RTD to Temperature] etc.) restored to their place – the export’s own summary list omits these bracketed headers and the now-unlabelled Manual Oil Valve rung, both noted below.

Step	Section
0	Timers
22	Load Once
43	E-Stops
56	Automation
65	Declarations
133	Analog to Digital
167	ADC Float
223	RTD to Temperature (P0)
266	Ultrasonic to Level (P1)
309	Flowmeter to RPM
336	Sensor Averaging, accumulate (P2)
366	Sensor Averaging, divide/reset (P3)
405	Level % Converter
562	Sensor Logging, 5 s snapshot (P4)
584	Rate-of-change ratio (P5)
614	Error Threshold (incl. trigger-value snapshot, D750–D768)
915	Water Temperature Target Logic
1038	Heater Latching
1210	Manual Heater Control
1220	Height to Mass
1273	Dynamic Upper Thermal Boundary
1349	PO Designation
1410	PO Timer (D302)
1587	PO Dispensing
1605	Tank Scoring (incl. manual tank disconnect override)
1760	Need Scheduler
1796	Scheduler Cases
1814	Mode Scheduler
1868	Mode Indicator
1892	LSM Scheduler
2058	HSM Scheduler
2364	FSM Scheduler
2538	Scheduler Gates
2582	Solar Gate
2607	Manual Solar Gate
2611	Oil Scheduler (MODBUS)
2843	Oil Distribution Calculation (“Oil Calculation”)
2946	Oil Distribution Gating
2959	Oil Distribution Timing (incl. unlabelled Manual Oil Valve rung, ~3012)
3015	Jacket Pump
3020	AODD Pump
3053	Manual Pump Mode (also carries the manual mode-select override)
3143	HMI Display for Inputs
3207	Oil Chart
3277	Charting
3298	END

5 Functional Walkthrough

5.1 Timer Chain (Step 0)

Unchanged from 3.2: **T0** ($K10 = 1$ s) power-up settling delay on M8000; **T1** ($K30 = 3$ s), gated by T0, the master “logic enabled” tick; **T2** ($K10 = 1$ s), gated by T1 and self-resetting, the free-running 1 s pulse driving P0/P1/P2; **T3** ($K50 = 5$ s), also self-resetting, driving P3/P4; **T4** ($K110 = 11$ s) driving P5 and Error Threshold. All Coolmay timers here run on a 100 ms base ($K10 = 10 \times 100$ ms = 1 s) – the same base behind the T250/T251 scaling in Section 5.10.

5.2 Load Once (Step 22) and Declarations (Step 65)

Unchanged in substance from 3.2, one step earlier throughout: first-scan zeroing of the daily oil-chart totals, and the always-on initialisation of Z0–Z5, D550, D303/D304, and the comm-port special registers (D8114, D8397, D8400, D8401, M8192).

5.3 E-Stops (Step 43) and Automation (Step 56)

Unchanged in substance. X000 AND M60 drive M10 (E-Stop condition, reading fault bits M0–M9), which gates Y000. T1 AND M10 AND X001 drive M11 (master auto-mode flag); Y001 AND X001 drive M12 (manual-mode selector). Y001 remains the automation-active indicator.

5.4 Analog to Digital, ADC Float, RTD/Ultrasonic/Flowmeter Scaling (Steps 133–336)

Unchanged in substance from 3.2, one step earlier throughout. RTD temperature, ultrasonic level and the flowmeter still share the identical placeholder scale/offset pair (E2.5, E1) – unchanged, and still carried in Section 9.

5.5 Sensor Averaging (Steps 336–404) and Level % Converter (Step 405)

Unchanged in substance: P2 accumulates all eleven channels once per second, P3 divides by 5 and resets every 5 s. The Level % Converter still runs the K100 subtract/divide/multiply round trip described in the 3.2 edition – normalise to a 0–1 fraction of the K100 placeholder span, then re-express against that same span – for all four level channels (T5, TA, TB, TC), landing in D122, D124, D126, D128.

5.6 Error Threshold (Step 614)

Gated by T4 (11 s), still ten comparisons of the rate-of-change ratios (D220–D238) against paired fault thresholds (D240–D278), driving fault bits M0–M9 – unchanged from 3.2. The flow channel (ratio register D240 in the Z5 band) is still not compared here.

New this edition: a trigger-value snapshot per channel. Each comparison rung carries a second, paired check that, when the rate-of-change value drops back below a second (lower) threshold register, moves that value into a dedicated snapshot register: D750 for channel 0 through D768 for channel 9. These D750-and-up registers are the values shown on the HMI as the “**max value recorded**” for each channel – a trigger-value snapshot capturing what the rate-of-change actually was at the moment the fault condition cleared, rather than only the live fault bit (M0–M9) with no record of how far it went.

5.7 Water Temperature Target Logic (Step 915)

Unchanged. T_{target} (D130) against Level T5% (D122): flat at 70.0 below 50%, flat at 40.0 above 90%, and the same quadratic ramp between:

$$D130 = 70 - \frac{3(D122 - 50)^2}{160}, \quad 50 \leq D122 \leq 90.$$

Both endpoints still close exactly (70.0 at D122=50, 40.0 at D122=90).

5.8 Heater Latching (Step 1038) and Manual Heater Control (Step 1210)

Gated by M11, this is a clean two-path split on the sign of D220 (Temp T5 rate-of-change), each path running its own threshold cascade against Level T5% (D122) into the stage counter D350:

Rising ($D220 > K1$): the top three thresholds set the stage – $D122 \leq 55 \rightarrow K3$, $D122 \geq 55 \rightarrow K2$, $D122 \geq 80 \rightarrow K1$, $D122 \geq 95 \rightarrow K0$ (four rungs covering the three transition points, with both directions of the 55 boundary written out so the cascade lands correctly regardless of scan order).

Not rising ($D220 < K1$ or $D220 = K1$, i.e. $D220 \leq K1$): the bottom three thresholds set the stage – $D122 > 90 \rightarrow K0$, $D122 \leq 85 \rightarrow K1$, $D122 \leq 75 \rightarrow K2$, $D122 \leq 50 \rightarrow K3$. The equal-to-K1 case is simply grouped with the falling case, sharing the same threshold set – there is no third, separate branch here.

D350 cascades into Y010/Y011/Y012 (heater stages 1–3) unchanged. Manual Heater Control (M12, X006/X007/X010) unchanged.

5.9 Height to Mass (Step 1220) and Dynamic Upper Thermal Boundary (Step 1273)

Unchanged in substance. Height-to-mass is still unity gain (K1), pending real site conversion figures (Section 9). The mixing formula for T_f (D296) still uses the E239.53 constant, unchanged, and the K55 boundary compare (setting M13) is unchanged.

5.10 PO Designation (Step 1349), PO Timer/D302 (Step 1410), PO Dispensing (Step 1587)

PO Designation is unchanged in substance. The two-branch melt/liquid demand function producing D302 is unchanged in its maths.

PO-dispense watchdog renumbered to a 100 ms device. 3.2 used the Coolmay device **T5** directly against D302. This export computes **D340 = D302 × E10** and gates a timer named **T250** (contacts: T0 AND M50 AND M51 AND D340) whose preset is D340 itself; on timeout, D340 is cleared (DMOV K0 D340) and, gated by M52, M50 is reset. Same watchdog-and-clear pattern as 3.2 – start condition, dynamic preset, clear-on-timeout, reset-on-completion – against a value now expressed in the timer’s native 100 ms units (D302 in seconds × 10 = deciseconds) rather than seconds directly.

5.11 Tank Scoring and Manual Disconnect Override (Step 1605)

Weighted score per tank = $0.6 \times \text{mass proxy} + 0.4 \times \text{temperature}$ (D330 TA, D332 TB, D334 TC) – unchanged, same weighting as 3.2. The manual disconnect coils M700/M701/M702 still gate the score computation and still force the corresponding score to zero elsewhere in the same section (around step 1730 onward) – unchanged; full description in Section 6.

5.12 Need Scheduler through Scheduler Gates (Steps 1760–2578)

Unchanged in substance from 3.2: the two-stage design (tank scores decide the macro-mode LSM/HSM/FSM; within the chosen mode, a decision cascade through the intermediate flags M100–M109 picks the exact Y011–Y014 output pattern) still holds, and the governing principle – favour maximum parallel tank activation below 70% T5 level, stepping down only when fewer tanks are actually available – is unchanged. A disconnected tank’s zeroed score still removes it from every comparison in this chain without a separate interlock.

5.13 Solar Gate (Step 2582) and Manual Solar Gate (Step 2607)

Unchanged: D130 vs. D72 (Temp HEA) routes to Y020/Y021, combined with Y013; M12 AND X011 gives the manual override.

5.14 Oil Scheduler / MODBUS (Step 2611)

Unchanged from 3.2’s free-for-all redesign: a station’s HMI start button SETs that station’s lock (M200/M201/M202), locking out the other two; ADPRW performs the MODBUS read/write-back via D550 and D360; a failed exchange leaves the lock engaged until a successful reset or manual intervention. T10/T11 (station 1), T12/T13 (station 2), T14/T15 (station 3), and T16 (write-back gate) all unchanged (K1/K2 presets).

5.15 Oil Distribution Calculation, Gating and Timing (Steps 2843–3014)

The MODBUS-read value (D360) and processed flow (D42) still combine through DEDIV into D306, and an adjustable rate parameter (D500) with a DEXP estimate still produce the dispense-duration value, D316.

Flowmeter-fitted toggle, M800. An HMI switch selecting whether a physical flowmeter is installed at this site. When set (flowmeter not fitted), DEMOVE1 D42 forces the processed flow register to a fixed value of 1, letting the Oil Distribution timing maths run off an assumed flow rate; when clear, D42 takes its normal value from the Flowmeter-to-RPM scaling step (Section 3).

Dispense timer renumbered to a 100 ms device. 3.2 loaded D316 directly as the preset for timer **T100**. This export multiplies D316 by E10 in place first (DEMUL D316 E10 D316), then loads it as the preset for **T251** – same role (gates Y005 for whichever station holds the lock, reset by the limit-switch inputs X014/X015/X016/X017/X020/X021), renumbered device, same 100 ms-base reasoning as T250.

Master Oil Controller enable, M65. Y005’s dispense rung is gated by **M11 AND M65 AND** the active station lock AND its start input (e.g. X014). M65 is the master Oil Controller enable, sourced from the **TK8100H** supervisory HMI – not from any of the three per-station TK8050H units – giving the TK8100H a system-wide dispense enable/disable that sits above the per-station lock.

A manual oil-valve rung (M12 AND X003 → Y005) is still present, immediately following the auto dispensing rungs around step 3012, but no longer carries its own section header in this export – functionally unchanged from 3.2’s “Manual Oil Valve” section, just unlabelled.

5.16 Jacket Pump (Step 3015), AODD Pump (Step 3020), Manual Pump Mode (Step 3053)

Jacket pump (Y006) still runs unconditionally on M11, manual override M12 AND X004, unchanged. AODD pump (Y007) compare is still against D66/D68/D70 (Temp TA/TB/TC), engaging when M11 is set and any tank temperature exceeds K30.

Manual Mode Scheduler folded into Manual Pump Mode. 3.2 carried the manual LSM/HSM/FSM override (M12, pushbuttons via M304–M306, driving Y002–Y004) as its own named section at step 1728, separate from Manual Pump Mode at step 2899. In this export only *Manual Pump Mode*, now at step 3053, appears as a named section – and it is the block driving both the manual mode-select outputs (Y002–Y004, M304–M306) and the pump-mode pushbuttons (M300–M303), consolidated under one heading as part of the ongoing segmentation clean-up. Both behaviours are unchanged, just under one name instead of two.

5.17 HMI Display for Inputs (Step 3143), Oil Chart (Step 3207), Charting (Step 3277)

All unchanged in substance from 3.2: the 32-point input mirror (X000–X037 into M115–M146), the K1440 (24-hour) rollover on D610 zeroing the daily per-station totals D604–D606, and the four-point HMI trend chart (D296, D64, D130, D76 into D600–D603) on every T0 pulse.

6 Manual Tank Disconnect Override

Unchanged from the 3.2 edition. If a tank is physically taken out of the process but the control logic has no way of knowing that, it will keep scoring and scheduling that tank as if it were still there; this override exists to stop that.

6.1 How it works

Three SET/RST buttons on the HMI, one per tank – **M700** (Tank A), **M701** (Tank B), **M702** (Tank C) – let me flag a tank as disconnected. Setting the relevant button forces that tank's weighted score to zero:

- M700 forces **D330** (Tank A's score) to zero.
- M701 forces **D332** (Tank B's score) to zero.
- M702 forces **D334** (Tank C's score) to zero.

Because the Need/Mode Scheduler family always picks the tank(s) with the highest score, a zeroed score can never win a comparison – the disconnected tank drops out of LSM/HSM/FSM selection on its own.

Same mechanism, unchanged. The score-gating coil (at the top of Tank Scoring, step 1605) and the zeroing rung (around step 1730 onward) for each tank use the same coil with opposite contact logic in each place, exactly as in 3.2 – normal contact where the score is computed, inverted where it is forced to zero.

6.2 Why this location in the logic

Unchanged reasoning from 3.2: the override sits at the score itself so every downstream comparison – Scheduler Cases, LSM/HSM/FSM, Scheduler Gates – inherits the exclusion automatically, and a disconnected tank cannot be re-selected by PO Designation either, since that block only acts on tanks that have already scored well enough to be in contention.

7 Timer/Counter Presets

Device	Preset	Role
T0	K10 (1 s)	Power-up settling delay
T1	K30 (3 s)	Master logic-enabled tick
T2	K10 (1 s)	Self-resetting sample pulse (P0/P1/P2)
T3	K50 (5 s)	Self-resetting average/log pulse (P3/P4)
T4	K110 (11 s)	Rate-of-change / fault-check pulse (P5, Error Threshold)
T10, T12, T14	K1	MODBUS sequencer, stations 1–3 (primary exchange)
T11, T13, T15	K2	MODBUS sequencer, stations 1–3 (secondary exchange)
T16	K2	Write-back exchange step gate (all three stations)
T250	D340, 100 ms base (=D302×10)	PO-melt-demand watchdog – renumbered from T5 (3.2)
T251	D316, 100 ms base (=D316×10 in place)	Oil dispense duration – renumbered from T100 (3.2)

8 Reconstructed Register Map

Table 1: Reconstructed Register Map – Data Registers (D)

Register Range	Inferred Role
D0, D2	Raw onboard analogue channels (temperature)
D4–D10	Raw Slot-1 (RTD) channels, floated
D12–D18	Raw Slot-2 (ultrasonic) channels, floated
D20	Raw flowmeter channel (from D8231), floated
D22–D32	Processed Temp T5/TA/TB/TC/HEA/SOL
D34–D40	Processed Level T5/TA/TB/TC
D42	Processed flow (RPM) – forced to K1 when the M800 flowmeter-not-fitted toggle is set
D44–D64	Averaging accumulator, all 11 channels
D64–D84	Final 5-sample average, all 11 channels (D64 Temp T5, D66 TA, D68 TB, D70 TC, D72 HEA, D74 SOL, D76 Level T5, D78 TA, D80 TB, D82 TC, D84 Flow)
D100, D110, D112	Scratch registers for RTD/ultrasonic/flow scaling
D114–D120	Level % Converter intermediate (fraction stage)
D122–D128	Level % Converter output, T5/TA/TB/TC
D130	T_{target}
D132–D138	T_{target} quadratic-ramp scratch
D200–D218	5-second-old snapshot (P4)
D220–D238	Rate-of-change ratio (P5)
D240–D278	Adjustable fault thresholds, 10 channels (paired upper/lower per channel)
D750–D768	Trigger-value snapshot per channel – the HMI “max value recorded” display, new this edition (Section 5.6)
D280, D282, D284, D286	Mass proxies, T5/TA/TB/TC (unity gain – calibration item)
D288–D296	Dynamic Upper Thermal Boundary (T_f) calculation
D298, D300	Selected-tank temperature / mass (PO Designation)
D302	PO melt/liquid demand value
D303, D304	MODBUS exchange counters
D340	D302×10 – preset for the renumbered PO-watchdog timer T250
D306–D316	Oil dispense-duration estimate chain (D316 scaled ×10 in place before use as the T251 preset)
D318–D334	Tank-score components and final scores (TA/TB/TC)
D350	Heater stage count, 0–3
D360	MODBUS-polled raw value
D400	Scratch, PO demand chain
D480	HMI mode-name string (“FSM”/“HSM”/“LSM”)
D500, D502	Dispense-rate / estimate parameters
D550	MODBUS write-back value to active satellite
D600–D603	HMI trend-chart mirror
D604–D607	Daily per-station oil-distribution totals
D610	Daily/period rollover counter
D8030, D8031, D8114, D8231, D8397, D8400, D8401	Coolmay special registers (analogue channel raw values, comm-port setup)

Table 2: Reconstructed Register Map – Internal Relays (M)

Register Range	Inferred Role
M0–M9	Sensor rate-of-change fault bits (Error Threshold, step 614), also read at the E-Stops rung (step 43) to decide a shutdown
M10	E-Stop condition (X000 AND M60)
M11	Master auto-mode / system-enabled flag
M12	Manual-mode selector
M13	Dynamic-boundary-exceeded flag (D296 > K55)
M14–M16	PO tank-select flags, TA/TB/TC
M17–M19	Need Scheduler intermediate flags
M20–M25	Scheduler mode/case latches
M50–M52	Initiate / Run-or-Pause / Reset button trio, PO Dispensing
M60	HMI override button – manual force-shutdown for a persistent fault
M100–M109	Scheduler Gates – shared output-gating states
M115–M146	Physical input mirror, X000–X037
M200–M202	Oil Distribution free-for-all lock
M203–M206	MODBUS exchange handshake flags
M300–M306	Manual pump-mode selector inputs / manual mode-select outputs – both now under the single “Manual Pump Mode” heading
M700–M702	Manual tank disconnect override, Tank A/B/C – forces that tank’s score (D330/D332/D334) to zero (Section 6)
M65	Master Oil Controller enable, sourced from the TK8100H supervisory HMI – gates Y005 alongside M11 and the active station lock (Section 5.15)
M800	HMI toggle: flowmeter fitted / not fitted at this site – forces processed flow (D42) to K1 when set (Section 5.15)

Table 3: Reconstructed Register Map – Outputs (Y)

Output	Role
Y000	E-Stop output
Y001	Automation-active indicator
Y002–Y004	Mode outputs (LSM/HSM/FSM selection)
Y005	Oil dispense valve, active station (also driven manually via M12 AND X003)
Y006	Jacket pump
Y007	AODD pump
Y010–Y012	Heater stages 1–3
Y011–Y014	Tank priority outputs (Scheduler Gates)
Y020, Y021	Solar gate, high/low side
Y022–Y024	Oil distribution station routing

9 Calibration / Placeholder Constants

Unchanged from the 3.2 edition – everything below is still a constant sitting in the ladder that stands in for a real, site-measured value.

Location	Constant(s)	What it needs to become
RTD to Temperature / Ultrasonic to Level / Flowmeter to RPM	E2.5 (scale), E1 (offset) – identical across all three sensor types	Per-channel raw-to-engineering scaling, to be set from each transducer’s own datasheet/site calibration.
Level % Converter	K100 (full-scale span, all four level channels)	The real full-scale span per tank; the normalise-then-rescale structure is deliberate so only this constant needs updating per channel.
Height to Mass	K1 (DEMUL gain, ×4 channels)	Currently unity gain – needs the real height-to-mass conversion figure per tank, from tank geometry and product density.
Water Temperature Target Logic	K50, K90 (level bands), K70, K40 (flat target bounds)	Site-measured level bands and target-temperature bounds for the quadratic ramp.
Heater Latching	K55, K80, K95 (rising) / K90, K85, K75, K50 (falling), K1 (rate-of-change rising/not-rising threshold)	Heater-stage transition points and rate-of-change sensitivity – confirm against real sensor noise and site staging preference.
PO Timer / D302	K30 (melting threshold)	Melting-point reference constant – check against the actual palm-oil product specification.
AODD Pump	K30 (temperature compare, D66/D68/D70)	Trigger threshold for AODD pump engagement – confirm against real tank temperatures.

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Reconstructed from the Coolmay L10S “MAIN” ladder export, 15-Sep-2026 print, superseding the
Program 3.2 edition of this guide.